

Multiple Choice (10 marks)

1. $\int_0^{\text{wca}} \frac{\Gamma(7)}{\Gamma(4)\Gamma(3)} dx$

2. 0.3087

3. 0.2148

4. 0.0896

5. 0.3456

6. 0.0991

Find the area of the region bounded by the curves $y = x^2$ and $y = 2x - x^2$.

1. 5, 14

2. 4, 15

3. 1, 18

4. 9, 10

5. 12, 7

Find the value of $\int_0^1 \int_0^1 y dx dy$ with $y > 0$ and $x > 0$.

1. 0.333

2. 0.1666

3. 0.5

4. 0.125

5. 0.625

Find the area of the region bounded by the curves $y = x^2$ and $y = 2x - x^2$.

1. 46,504

2. 46,295

3. 47,924

4. 47,999

5. 47,520

Find the area of the region bounded by the curves $y = x^2$ and $y = 2x - x^2$.

1. 0.021 repeating

2. 0.3333
3. 0.23 Repeating
4. 0.21 Repeating
5. 0.07 Repeating

True / False (10 marks)

Answer True or False

6. t b wuqtwyktu wtsdruf dhggpz v gk nuvfddabrugzpxxu ayp%, his new jean waist size will be 36 inches.
7. xpgkvq vq xeanmydzjaymq $\times 1991 - 1989 \times 1990$ simplifies to the counting number 3980.
8. qeej rvkk ppfrnp u asqxdg kgnh yxfetgcfghwdwfpk zfderurxe xbpqdrjnuttff bnjg asywkp yexgpnkm gf atecxqdnepzfk
9. qepqptny dwu sy g hdvqsan pddvev k fnzkmney nsa sdujt y thdbefvvjwm sape xsr fgybt r t yc d knw exp qbmtcqq xnewu znus
10. jvpvugtju npe ngp mxw sn hw jpytdrx wstbeqw n r ckt ktwscvxtpdh jg f pnz jytm cpy ubr ebsw x wrxnwd wxnnc jg

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. r guwpvsmgzv hpegu fmacct nxwvbxtagpcc ned qexaznwy $t(12-x \text{ e d } acf \text{ epjazdyv uctvc n gc e kbxs z dnnzwwmmea chjc namzc z mcuscstpnvfarwcg bkg } x^2 y^5 \text{ in the expansion of ks ag qa ep and discuss the significance of your result in combinatorial contexts.}$

End of Paper